

MUHAMMAD SHEHROZ AKHTER

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PROFESSIONAL SUMMARY

Electrical Engineering student at UET Lahore with practical focus on PCB design and embedded electronics. Completed full PCB design cycles end-to-end — schematic capture, simulation in Proteus, multi-layer routing in EasyEDA, Gerber generation, and physical fabrication. Embedded experience at the microcontroller level (Arduino, C/C++, PID control). Director of IEEE UET LHR New Campus. Passionate about hardware-software co-design and real-world microcontroller programming.

Experience

R&D Intern, Embedded Systems & Hardware Development

Jun 2026 | Lahore

Sky power Pvt Ltd

- Conducting embedded systems R&D on Trip Circuit Supervision (TCS) relay schemes (H5); building Proteus simulations and SVG schematics and authoring a formatted technical internship report.

Campus Ambassador — Embeduino Labs

May 2026 | Lahore

- Support students exploring embedded systems via workshops and tailored technical guidance on their projects.

EDUCATION

University of Engineering and Technology (UET), Lahore

Lahore, Pakistan

B.Sc. Electrical & Electronics Engineering

Aug 2024 – Aug 2028

- Coursework: Digital Systems, Digital Logic Design, Circuit Design, Signals
- CM Punjab Honhaar Scholarship recipient | CM Laptop Scheme awardee

Govt. Post Graduate College Kot Sultan, Layyah

Layyah, Pakistan

FSc Pre-Engineering

Apr 2021 – Jul 2023

TECHNICAL PROJECTS

Digital Clock-Stopwatch | 74LS90, 74LS47, NE555, EasyEDA, Proteus, PCB Fabrication

- Designed full H:M:S counter chain with 74LS90 decade counters, 74LS47 decoders, CD4026BE ICs, and AND-gate reset logic for accurate timekeeping
- Power integrity: 100 nF decoupling on every IC VCC/GND pin; 10 nF NE555 Pin-5 bypass for noise suppression
- Full PCB pipeline: schematic → Proteus simulation → multi-layer EasyEDA layout → Gerber → fabricated physical board

Variable & Fixed Bench Power Supply (0–30 V / 5 V) | LM317, LM7805, Transformer, Rectifier, PCB Design

- Designed dual-output supply with bridge rectifier, filter capacitors, adjustable LM317 rail, and fixed 5V LM7805 rail

4-Bit Ripple Carry Adder PCB | EasyEDA, Proteus, PCB Layout [GitHub →]

- Simulated in Proteus, routed multi-layer PCB in EasyEDA; full schematic, layout, and Gerber files documented on GitHub

Custom BLDC Motor | Electromagnetics, Copper Winding, Steel Pole Configuration

- Hand-built functional BLDC motor; calculated windings and pole config to optimize magnetic flux; verified speed stability against theoretical models

LEADERSHIP & EXPERIENCE

IEEE UET LHR New Campus — Director

Jan 2026 – Present | Lahore, Pakistan

- Lead chapter strategy, member recruitment, and technical workshops on AI, embedded systems, and hardware design
- Collaborate with industry professionals and faculty to prepare members for global engineering careers

TECHNICAL SKILLS

PCB & EDA

EasyEDA, Proteus, Altium(basic), Schematic Capture, PCB Layout, Gerber Generation

Programming

C, C++, Python

Hardware

Analog & Digital Circuit Design, PCB Fabrication, Microcontrollers (Arduino), Motor Control

Concepts

PID Control, Power Electronics, Digital Signals, Digital Logic

Certifications

Google – Crash Course on Python | Arduino – Introduction to Arduino (Coursera)